

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 13: Mathematical Morphology

PART 1: FILL IN THE BLANKS

- Erosion **shrinks (or thins)** objects.
- Dilation **expands (or thickens)** objects.
- Opening = **Erosion** + **Dilation**
- Closing = **Dilation** + **Erosion**

PART 2: QUESTION & SOLUTION

How does morphology help in removing small noise?

Mathematical morphology uses combinations of erosion and dilation (such as Opening and Closing operations) based on a structuring element of a specific size:

- **Opening (Erosion followed by Dilation):** Removes small foreground noise points (salt noise) because objects smaller than the structuring element are completely eroded away, while larger objects are subsequently restored to their original dimensions via dilation.
- **Closing (Dilation followed by Erosion):** Eliminates small background holes or gaps (pepper noise) within foreground regions by bridging small breaks and subsequently restoring outer boundaries.